HyunSeo Jung

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Research Interests

Robotics, Automation, Control theory, Artificial Intelligence

Education

Pohang University of Science and Technology (POSTECH) — Pohang, South Korea
Bachelor of Science in Mechanical Engineering and Electrical Engineering; GPA: 4.02/4.30 — *Feb. 2020 – Feb. 2024*

- **Thesis**: Development of a Dynamic speed bump using emergency locking retractor mechanism
- **Thesis**: Reinforcement learning based Chip placement automation

Busan Science High School — Busan, South Korea
High School Diploma — *Mar. 2018 – Jan. 2020*

Publications

[C1] **H. S. Jung**, J. W. Kim, M. J. Kim, and S. H. Kang, "강화학습 기반 Chip Placement에서의 보상함수 설정 방법", The 30TH Korean Conference on Semiconductors, 2023

Honors and Awards

Best Capstone Design Award (Dept. of M.E., POSTECH) — 2023

- Awarded to a *POSTECH M.E.* student who presented outstanding research results in the capstone design project

Best Student Award (Dept. of M.E., POSTECH) — 2022

- Awarded to a *POSTECH M.E.* student who spent an excellent semester

Haesuk Cultural Scholarship (Haesuk Scholarship Foundation) (₩8,000,000 ≈ $6,400) — 2022-2023

Undergraduate Group Research Program (POSTECH) — 2021

- Topic: Development of a lipreading device
- 3rd prize

POSTECH Programming Contest — 2021

- 2nd prize

POSTECH Presidential Fellowship (POSTECH) (about ₩40,000,000 ≈ $32,000) — 2020-2024

- Full ride program for undergraduate course

Hanseong Fellowship (Hanseong Sonjaehan Scholarship Foundation) (₩10,000,000 ≈ $8,000) — 2019

- Awarded to academic excellent students

Research Experiences

Research Intern (MARCH Lab, POSTECH) — Jun. 2023 - Nov. 2023

- Research on balancing control of an unmanned unicycle

Research Intern (MARCH Lab, POSTECH) — Jun. 2022 - Aug. 2022

- Research on balancing control of an unmanned bicycle

Freshman Research Intern (Dept. of E.E, POSTECH) — Sep. 2020 - Dec. 2020

- Practice PID control using an inverted pendulum and 2-dof helicopter system

Freshman Research Intern (Dept. of M.E, POSTECH) — Sep. 2020 - Dec. 2020

- Practice PID control using a segway system

Teaching Experiences

Student Mentoring Program (Dept. of M.E., POSTECH) 2023

- Served as a tutor to teach Systems Control course

Teaching Assistant (K-MOOC) 2021-2022

- Developed course materials and assisted course operation for K-MOOC 'Introduction to quantum physics and computing' lecture

Seminar Host (PoApper) 2021

- Served as a mentor for frontend seminar at PoApper, the POSTECH Web/App development group

Extracurricular Activities & Miscellaneous

Chief Financial Officer (Undergraduate Association, POSTECH) 2023

Student President (Dept. of M.E., POSTECH) 2022

President of POSTECH Table Tennis club 2021-2022

Member of POSTECH Cheerleading team 2020-2021

Skills

- **Languages**: Korean(native), English(intermediate - TOEIC 920)
- **Hardware**: 3D printer, Lathe, Milling machine, Arduino, Raspberry pi
- **Software**: (including PyTorch and PyBullet framework), Matlab, C/C++, CAD(Solidworks & Autodesk Inventor)